ID: . Assessor(s): .

Structured Feedback

Design Debrief Axes of Evaluation

N = in range; U(L) = Developing (Low); D(L, M, H) = Developed (Low, Middle, High); Q = High Quality

		Development Level							
Component	**Elements**	**N**	**U**	**UH**	**DL**	**DM**	**DH**	**Q**	**Comments**
Design and Teamwork Considerations	Coherence across the Design Chain	☐	☐	☐	☐	☐	☐	☐	
	Utility of Prototype System in the context of ESC204	☐	☐	☐	☐	☐	☐	☐	
	Utility of Prototype System to investigate the Design Concept	☐	☐	☐	☐	☐	☐	☐	
	Analysis of the Teamwork Process	☐	☐	☐	☐	☐	☐	☐	
Design Debrief Approach Considerations	Quality of Engineering Argument	☐	☐	☐	☐	☐	☐	☐	
	Effectiveness of Teamwork during the Debrief	☐	☐	☐	☐	☐	☐	☐	
	Meaningful, Professional-style Communication	☐	☐	☐	☐	☐	☐	☐	

Holistic Score

Quality Score X.X / 10

Adjusted Score N/A / 10

Holistic Comments

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Feedback Guide

Criteria	Not Developed (N)	Developing (U)	Developed (D)	High Quality (Q)
Coherence across the Design Chain	From the Debrief discussion, it is unclear how the team progressed through the spaces on the Design Chain to develop the Design Concept and Prototype System based on their interpretation of the provided opportunity.	The team presents the Prototype System and opportunity context as well as their approach to the presented challenge and the Design Concept they developed; however, the connections between these elements may be unclear from discussion during the Debrief. Contextual factors considered may be based on assumptions, or feel like an afterthought to the technical design process. Connections to stakeholder needs may be vague.	The team describes how the design context connects to their Design Concept and Prototype System through the approach taken. The full design process from context to Prototype System demonstrates a strong understanding of the challenge presented in the chosen RFP. Stakeholder needs and other relevant contextual factors are clearly elaborated with reference to appropriate evidence from secondary sources.	*"Developed"* + connections across the Design Chain demonstrated during the Debrief conversation justify design outcomes (Design Concept, Prototype System), and strongly and seamlessly link them to the opportunity context. Stakeholder needs are clearly identified and centered in the design process.
Utility of Prototype System in the context of ESC204	The Prototype System violates one or more course constraints; OR The extent of prototyping activities (i.e., outcomes) is unclear from the Debrief conversation alone.	The Prototype System (or aspect(s) of it) is demonstrated (live or in a recording). The Prototype System appears to meet course constraints; however, discussion of the design, build and verification processes may be too vague or too detailed to effectively communicate the key features of the prototype system, milestones in development, and/or lessons learned throughout Phase 3.	The Prototype System (or aspect(s) of it) is demonstrated (live or in a recording), including response to various inputs/environmental factors/etc. as appropriate. The basic functionality of the Prototype System as-built is clear from the demonstration. The Prototype System and the process of developing it are discussed with enough detail to determine key features, milestones, and lessons learned in design, implementation, systems integration and testing during Phase 3.	*"Developed"* + Discussion of the Prototype System reveals a systematic and considered approach taken by the team to design, build and test a functioning Prototype System, which clearly meets course requirements. Key lessons learned throughout Phase 3 are highlighted, both for the team overall, and for individual members.
Utility of Prototype System to investigate the Design Concept	The utility of the Prototype System to investigate one or more aspects of the Design Concept is unclear from the Debrief conversation alone.	Some attempt is made to connect the Prototype System to the Design Concept; however, rationale may be vague or lack sufficient justification to strongly motivate the Prototype System as a means of demonstrating, investigating, or evaluating relevant aspect(s) of the Design Concept.	Sufficient rationale is provided to connect the Prototype System to the Design Concept (through a developed Specification, requirements flow-down, or other evidence), which makes the utility of the Prototype System in demonstrating, investigating, or evaluating relevant aspect(s) of the Design Concept clear.	*"Developed"* + Evidence provided shows that the Prototype System fulfills its purpose(s), and clearly demonstrates something meaningful about the Design Concept.
Analysis of the Teamwork Process	Discussion of teamwork is vague, or account/analysis of the teamwork process is not coherent between team members.	Discussion of teamwork highlights project management strategies used and the plan for the creation of the Prototype System; however, analysis of their effectiveness may be lacking. Discussion may not include identification of practices the team (members) will/will not take forward for future teamwork.	Discussion of teamwork integrates evidence which highlights the project management strategy(ies) used during Phase 3. The team's overall plan for creation of the Prototype System is discussed and analyzed in terms of its effectiveness, including challenges faced during implementation/integration. Specific and actionable future practices are offered and connected to the discussion.	*"Developed"* + the team's project management strategies and overall plan for the Prototype System are meaningfully appraised by the team, which supports future practices offered; analysis/appraisal of any strategies used and their effectiveness is offered with connections to team (member) identity and other team factors.
Quality of Engineering Argument	No or minimal evidence from the Design Dossier (or otherwise) is referenced during the Debrief. Many claims made by the team appear to be unsubstantiated.	An attempt is made to justify claims made during the Debrief with reference to relevant evidence from the Design Dossier; however, the selected pieces of evidence do not strongly support the discussion, interrupt discussion significantly, or are not easily interpretable by the audience.	The team generally provides strong justification for claims made during the Debrief using evidence from the Design Dossier. When used, evidence is integrated into the discussion and does not distract from the flow of the conversation.	*"Developed"* + evidence is effectively and seamlessly integrated into most aspect(s) of the conversation, including the 20-minute Q&A. Evidence is presented with particular clarity that makes it easily interpretable by the audience.
Effectiveness of Teamwork during the Debrief	One team member dominates the conversation by talking over others, or not creating space for other team members to share perspectives; OR Team members talk over, contradict, or argue with each other.	Each team member contributes to the Debrief, either through the 15-Minute Overview, the conversation, or both; however, when attempting to add to the conversation, each individual may contribute separate information or perspectives without building on other team members' contributions.	All members of the team contribute meaningfully to the Debrief. When appropriate, each individual builds on points made by teammates in a constructive and respectful manner.	*"Developed"* + team members demonstrate active listening towards their teammates, and are able to provide additional interpretations, follow-up comments, and other points which extend information/perspectives offered.

ID: .

Assessor(s): .

Criteria	Not Developed (N)	Developing (U)	Developed (D)	High Quality (Q)
Meaningful, Professional-style Communication	Language used by the team may be vague, evasive, or excessively colloquial. Use of jargon may impede clarity. The conversation is confrontational, argumentative, or defensive in tone, which detracts from the team's ability to communicate design and teamwork completed.	In general, language used throughout the Debrief is appropriate for a professional, engineering setting. Excessive use of jargon may impede understanding for the audience at some isolated points in the conversation, but not throughout. Discussion of design work generally demonstrates respect towards stakeholders and their lived experiences. The team does attempt to respond to the prompts provided; however, sometimes answers are vague, rambling, or oblique to the question asked.	Language used throughout the Debrief is appropriate and professional. Use of jargon (if present) is appropriately defined such that it does not impede communication. Discussion of design work demonstrates respect towards stakeholders and their lived experiences and has a positive tone. In general, responses fully address prompts/questions from the Teaching Team with minimal rambling.	*"Developed"* + team demonstrates active listening behaviour when engaging with the Teaching Team, and provides thorough but succinct responses to questions asked.